

Aaron Ma

Department of Computer Science ◊ University of Alberta ama11@ualberta.ca

Research interests: Theoretical reinforcement learning, spectral graph methods for exploration, representation learning, sequential decision-making

EDUCATION

Masters of Science in Computer Science September 2025 - Present
University of Alberta, Supervised by Professor Marlos C Machado, *Fully funded thesis based*

Bachelors of Science with Honors in Computer Science September 2021 - June 2025
University of Toronto (St. George Campus), GPA: 3.83/4.0, *Graduated with high distinction*
Selected coursework: MAT257 (Honors Analysis II), MAT354 (Complex Analysis), MAT347 (Abstract Algebra), CSC265 (Enriched Data Structures and Algorithms), CSC412 (Probabilistic ML) APM461 (Combinatorial Methods)

EXPERIENCE

MSc. at University of Alberta and AMII September 2025 - Present
Supervisor: Professor [Marlos C Machado](#). *Edmonton, AB*

- Investigated spectral graph-based option discovery methods in reinforcement learning, with a focus on eigenoptions and long-horizon exploration
- Developed a modification of the eigenoptions framework based on alternative Laplacian constructions, improving empirical l1-coverage
- Conducted large scale evaluation of eigenoptions with respect to the L1-coverage metric, informing ongoing theoretical analysis
- Currently deriving regret bounds linking spectral gaps in the environment graph to sample efficiency of reinforcement learning algorithms (working towards publication)

Vector Institute, University of Toronto August 2024 - Present
Supervisor: Professor [Roger Grosse](#) *Toronto, ON*

- Designed compact transformer-based parameterizations for twist functions, using attention mechanisms to model conditional dependencies in latent-variable inference
- Developed a contrastive training pipeline for latent-variable models using Sequential Monte Carlo

WATonomous, University of Waterloo May 2024 - August 2024
Supervisor: Dr. Mohammad Al-Sharman *(Remote) Waterloo, ON*

- Implemented a modified Soft Actor-Critic algorithm in PyTorch for off-policy continuous control using human demonstrations
- Resolved gradient-instability bugs and optimized numerical routines, improving convergence speed and stability

Classification of Lie Algebras, University of Toronto March 2023 - September 2023
Supervisor: Professor Joseph Repka *Toronto, ON*

Research topics: Symbolic computation, Lie Algebras and Representation Theory, Combinatorics.

- Developed and optimized a high-performance symbolic computation package in C++ to compute key structural properties of matrix algebras
- Improved performance in critical subroutines through increasing parallelism and cache-locality; optimizations achieved a **3x** speedup

Algebraic Topology and Game Theory Research, University of Toronto September 2022 – July 2023
Supervisor: Dr. Malors Espinosa Lara *Toronto, ON*

- Analyzed strategic equilibrium and winner determination in the game of Hex on non-planar surfaces (torus, Klein bottle)
- Proved a topological correspondence relating simplicial and covering constructions establishing unique-winner properties
- Applied algebraic topology techniques to reason about winner determination in two-player combinatorial games
- Presented findings at the **Canadian Undergraduate Mathematical Conference (CUMC) 2023**

Teaching Assistant at University of Alberta September 2025 - December 2025
Course: CMPUT 481 - Parallel and Distributed Systems Coordinator: Professor Paul Lu *Edmonton, AB*

- Developed an Optical Character Recognition (OCR) pipeline combining OpenCV for document preprocessing and a PyTorch-based recognition model for test marking. Reducing grading time by 80%.
- Prepared and delivered tutorial on tensor parallelism implementation using MPI

PROJECTS

Proximal Policy Optimization Implementation

- Implemented Proximal Policy Optimization from scratch using **Pytorch**.
- Achieved mean score of **8000** in 10m steps on the MuJoCo Humanoid environment. (competitive with stable base-lines/cleanrl).
- Designed a **mechanistic interpretability pipeline** to analyze one step optimization dynamics; fixed minibatch sequence and identically seeded optimizers for both the weight spaced perturbed actor and
- Quantified policy divergence by measuring **cosine similarity in weight space**, characterizing the relationship between parameter-space noise and the stability of the trust-region update.
- Optimized training throughput by **2x** on consumer-grade hardware (GTX 1070) by hiding CPU/GPU communication latency through non-blocking memory transfers and overlapping compute with data movement.

Quantitative MMA Prediction Modeling

May 2025 - Present

Independent Software engineering and Machine Learning research

- Modeled UFC fight outcomes as a probabilistic pricing problem, treating betting odds as implied market probabilities
- Designed models to identify and exploit systematic mispricings in published odds under uncertainty
- Built an end-to-end research pipeline including data ingestion, feature construction, model training, and walk-forward backtesting
- Evaluated strategies using risk-adjusted performance metrics and robustness checks over historical betting markets

KidneyOS, University of Toronto

May 2024 - August 2024

Software Developer, supervised by Professor [Jack Sun](#)

Toronto, ON

- Led a team of three developers, implemented virtual file system and IDE device drivers for KidneyOS (a Rust-based OS designed for teaching)

AWARDS & HONORS

- **Canadian Mathematical Olympiad 2021 Qualifier**: Invited as one of the top 89 COMC participants nationwide
- **Euclid Mathematics Contest 2021**: Achieved top 50 out of 20,000 contestants
- **Dean’s List Scholar, University of Toronto**: Recognized in 2022, 2023, 2024 and 2025 for academic excellence
- **University of Alberta Graduate Recruitment Scholarship**: Received \$5000 for academic excellence

SKILLS

Languages	C, C++, Rust, Python, Java, Assembly, Javascript, SQL, HTML, CSS. Haskell, Go
Libraries/Frameworks	Pytorch, Jax, Tensorflow, Scikit-Learn, Numpy, pandas, CUDA, OpenMPI, CVXPY, Sqlite
Tools	Git, JUnit, Regex, UNIX/Linux, Nix, Docker, Docker-Compose, Kubernetes, Proxmox-VE,